

Mini Project Report

**Restoration of Fog/Haze-Degraded Outdoor Images
Using Dark Channel Prior and Guided Filtering**

1. Title

Restoration of Fog/Haze-Degraded Outdoor Images Using Dark Channel Prior and Guided Filtering

2. Problem Statement

Images of outdoor scenes captured in foggy or hazy atmospheric conditions suffer from reduced contrast, washed-out colours, and loss of fine detail. This degradation is caused by the scattering and absorption of light by suspended atmospheric particles (fog, mist, dust, smoke) before it reaches the camera sensor. As a result, the light actually received by the camera is a blend of the true scene radiance and scattered ambient light (airlight), which visually manifests as a hazy, low-contrast image.

This project addresses the following restoration problem: given a single hazy/foggy RGB image, recover an estimate of the original, haze-free scene as faithfully as possible, without any additional sensor information (such as depth or multiple exposures). This is a classic single-image restoration problem because it requires inverting a physical image-degradation model (the atmospheric scattering model) to recover the latent clean image from one degraded observation — directly matching the restoration theme of the assignment (CO4).

3. Objectives

- To study the atmospheric scattering (haze formation) model that describes how fog degrades an outdoor image.
- To generate a controlled, synthetic hazy image from a clean photograph using the atmospheric scattering model, so that a ground-truth clean image is available for objective evaluation.
- To implement the Dark Channel Prior (DCP) algorithm to estimate the atmospheric light and the scene transmission map directly from the hazy image.
- To refine the raw transmission map using a manually implemented guided filter, so that the map respects object edges instead of the blocky boundaries produced by the raw dark-channel estimate.
- To recover the haze-free image by inverting the atmospheric scattering equation using the refined transmission map and estimated atmospheric light.
- To quantitatively evaluate the quality of restoration using PSNR and SSIM, comparing the hazy image and the dehazed image against the known clean reference.
- To analyse the limitations of the Dark Channel Prior approach and suggest directions for improvement.

Relevance of the Problem

Haze removal is directly useful wherever outdoor imagery drives a downstream decision. Surveillance and traffic-monitoring cameras lose object visibility in fog; autonomous-driving and ADAS perception pipelines suffer accuracy drops on hazy roads; aerial/satellite imagery and drone photography are frequently degraded by atmospheric haze; and everyday outdoor photography benefits from contrast and colour restoration. Because a single degraded image is often all that is available (no stereo pair, no depth sensor), a physically motivated, prior-based restoration method such as the Dark Channel Prior is a practical and interpretable solution — it needs no training data and works on a single image, unlike most deep-learning dehazing networks.

4. Dataset / Image Description

- **Source:** A single clean outdoor photograph (an urban skyline scene, file: photo1.jfif) was used as the ground-truth clean image.
- **Number of images:** 1 base image. From this single clean image, one synthetic hazy version was generated, giving a clean–hazy–dehazed image triplet for evaluation.
- **Image characteristics:** Colour (RGB) outdoor cityscape photograph containing buildings, sky, and a clear horizon — a scene type well suited to the Dark Channel Prior, since the sky region and the building façades provide contrasting regions for testing the atmospheric-light estimate.
- **Why synthetic haze:** Real hazy/clean image pairs of the exact same scene are difficult to obtain, since haze density changes constantly and a truly haze-free reference of the same moment rarely exists. To allow quantitative, ground-truth-based evaluation (PSNR/SSIM), haze was synthetically added to the clean photograph using the standard atmospheric scattering model: $I_{\text{hazy}} = I_{\text{clean}} \times t + A \times (1 - t)$, with a fixed atmospheric light $A = [0.9, 0.9, 0.9]$ and a uniform transmission $t = 0.55$ (moderate, uniform haze density).
- **Preprocessing:** The image was read with OpenCV, converted from BGR to RGB colour order, and normalised from 8-bit integers to floating-point values in the range $[0, 1]$ before any further processing.
- **Limitations of the image data:** Only one base scene was used, and the haze was synthetically generated with a single, spatially uniform transmission value. Real-world fog is rarely perfectly uniform and the true atmospheric light is rarely a flat grey; this is discussed further in the Limitations section.

5. Methodology

The overall processing pipeline follows the stages below:

Clean Input Image → *Synthetic Haze Generation (Atmospheric Scattering Model)* → *Dark Channel Computation* → *Atmospheric Light Estimation* → *Raw Transmission Map Estimation* → *Transmission Map Refinement (Guided Filter)* → *Scene Radiance Recovery (Dehazing)* → *Result Comparison* → *Quantitative Evaluation (PSNR, SSIM)*

Stage 1 — Synthetic Haze Generation

A clean photograph is loaded and normalised to $[0, 1]$. Haze is synthesised using the atmospheric scattering (Koschmieder) model, $I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x))$, where J is the clean scene radiance, A is the atmospheric light, and t is the transmission. A fixed $A = [0.9, 0.9, 0.9]$ and $t = 0.55$ were used to create a controlled, uniformly hazy version of the image, which also serves as the ground truth for later evaluation.

Stage 2 — Dark Channel Computation

For haze-free outdoor patches, at least one colour channel typically has very low intensity in some pixels (shadows, coloured surfaces, dark objects) — this is the Dark Channel Prior. The dark channel is computed by taking the minimum across the R, G, B channels at every pixel and then applying a minimum filter (implemented here as morphological erosion with a 15×15 structuring element) over a local neighbourhood. In hazy regions, the dark channel has consistently high intensity, which is what makes it useful for detecting and quantifying haze.

Stage 3 — Atmospheric Light Estimation

The brightest 0.1% of pixels in the dark channel are assumed to correspond to the most haze-opaque region of the scene (typically sky or distant background). Among these candidate pixels, the maximum RGB value in the original hazy image is taken as the estimate of the atmospheric light A .

Stage 4 — Transmission Map Estimation and Refinement

A raw transmission map is estimated from the dark channel of the image normalised by the atmospheric light, using $t(x) = 1 - \omega \cdot (\text{dark channel of } I/A)$, with $\omega = 0.95$ (retaining a small amount of haze for distant objects to preserve depth cues). Because this raw estimate is blocky (it inherits the patch-based minimum-filter structure), it is refined using a guided filter, implemented manually via local linear regression with box filters (mean, variance and covariance computed over a 40×40 window with a regularisation $\epsilon = 0.001$), using the grayscale hazy image as the guidance image. This produces an edge-aware transmission map that follows object boundaries instead of the coarse block structure of the raw estimate.

Stage 5 — Scene Radiance Recovery

The haze-free image is recovered by inverting the scattering model: $J(x) = (I(x) - A) / \max(t(x), t_0) + A$, with a lower bound $t_0 = 0.1$ imposed on the transmission to avoid noise amplification in regions where the estimated transmission is very close to zero. The result is clipped to a valid $[0, 1]$ range and saved as the dehazed output.

Stage 6 — Comparison and Evaluation

The hazy image, dark channel, raw and refined transmission maps, dehazed result, and original clean image are displayed side by side for visual comparison. PSNR and SSIM are then computed between (a) the hazy image and the clean image, and (b) the dehazed image and the clean image, to quantify the improvement achieved by the restoration pipeline.

6. Implementation

- **Language:** Python 3
- **Libraries used:** OpenCV (cv2) for image I/O, colour conversion, morphological erosion and box filtering; NumPy for array/matrix operations; Matplotlib for visualisation; scikit-image (skimage.metrics) for PSNR and SSIM computation.
- **Core techniques implemented:** Atmospheric scattering (haze) model, Dark Channel Prior, top-percentile atmospheric light estimation, box-filter-based guided filtering (implemented from first principles rather than using a library call), and inverse-model scene radiance recovery.
- **Key parameters:** Dark-channel patch size = 15×15 ; haze retention $\omega = 0.95$; guided-filter radius = 40, $\epsilon = 0.001$; minimum transmission $t_0 = 0.1$.

The guided filter was deliberately implemented manually (rather than calling `cv2.ximgproc.guidedFilter`) using box-filter statistics, so that the underlying local-linear-model mathematics of edge-preserving smoothing was fully understood and explainable rather than treated as a black box, in line with the self-study expectation of this assignment.

7. Results

The figure below shows (left to right) the dark channel of the hazy image, the synthetically hazy input image, and the final dehazed result produced by the pipeline, captured directly from the project's execution output.

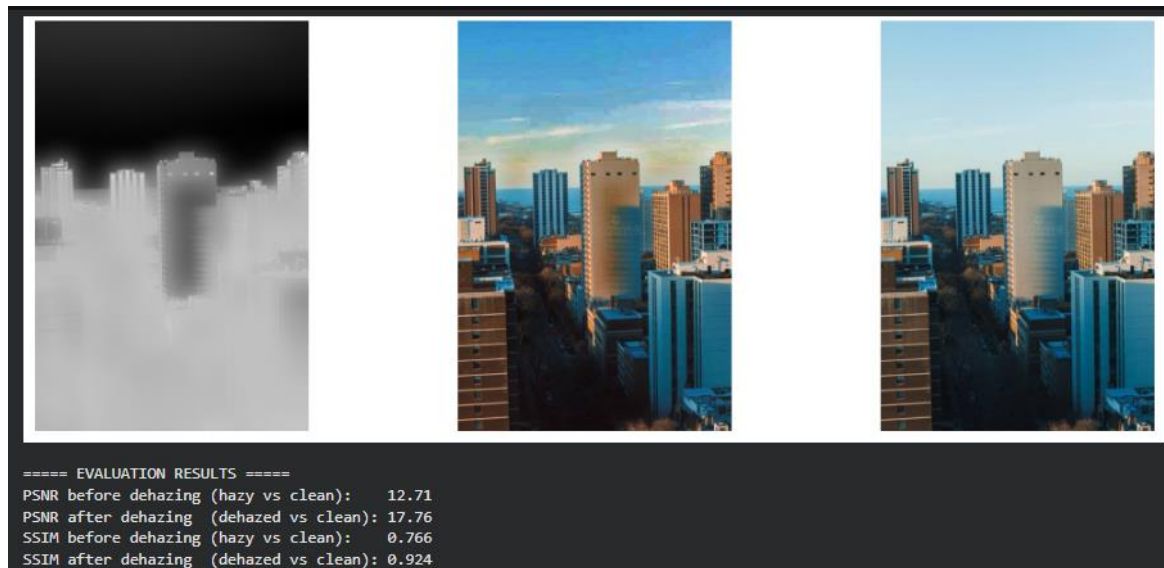


Figure 1: Dark channel, synthetically hazy input, and dehazed output (with printed evaluation metrics).

Visually, the dehazed result shows a noticeably clearer sky, higher contrast on the building façades, and colours that are much closer to the original clean image than the hazy input — confirming that the recovered transmission map correctly identified distant/hazy regions (upper sky) versus nearer, less-hazy regions (buildings).

Quantitative Evaluation

Metric	Hazy vs Clean (Before)	Dehazed vs Clean (After)
PSNR (dB)	12.71	17.76
SSIM	0.766	0.924

The restoration pipeline improved PSNR by about 5.05 dB (from 12.71 dB to 17.76 dB) and improved SSIM from 0.766 to 0.924, an increase of roughly 0.158 (about 21% relative improvement in structural similarity). Since PSNR is sensitive to overall pixel-intensity error and SSIM captures structural/perceptual similarity, the joint improvement in both metrics indicates that the Dark Channel Prior + guided filter pipeline restored both the correct brightness/colour balance and the structural detail (edges, textures) that were degraded by haze, rather than improving one at the expense of the other.

8. Limitations

- Evaluation was performed on a single synthetically hazy image; performance on real fog/haze photographs with non-uniform, spatially varying haze density has not been tested.
- The atmospheric light was assumed to be a single flat RGB value across the whole image (both in the synthetic haze generator and in the estimation stage), which does not hold for scenes with artificial lighting, coloured haze, or non-uniform illumination.

- The Dark Channel Prior assumption (at least one channel is near-zero in a local haze-free patch) can fail in large sky regions or bright, near-white surfaces, which can lead to colour shifts or halo artefacts around high-contrast edges — a known weakness of DCP-based methods.
- Key parameters (ω , guided-filter radius/ ϵ , t_0) were fixed manually rather than tuned adaptively per image; results may be sensitive to these choices for images with different haze density or scene content.
- The method has not been tested on dense fog, night-time haze, underwater imagery, or smoke, where the atmospheric scattering assumptions used here are known to be less accurate.
- The guided filter, dark-channel erosion and per-pixel operations are computationally more expensive on very high-resolution images; runtime was not benchmarked in this project.

9. Future Scope

- Evaluate the pipeline on a larger, more diverse set of real hazy-image benchmarks (e.g., RESIDE, O-HAZE, I-HAZE) that include genuine hazy/clean image pairs, rather than only synthetic haze.
- Compare the classical Dark Channel Prior approach against learning-based dehazing methods (e.g., AOD-Net, FFA-Net, or transformer-based dehazing models) to quantify the gap between prior-based and data-driven restoration.
- Introduce adaptive/automatic parameter selection (e.g., data-driven ω or guided-filter radius) instead of fixed constants.
- Add a segmentation stage (e.g., separating sky from foreground) so that the atmospheric light and transmission estimation can be handled differently for different scene regions, addressing the halo-artifact limitation.
- Extend the project to real-time video dehazing and explore deployment as a lightweight web or mobile application.
- Use ground-truth hazy/clean datasets to also report Dice/IoU-style region consistency and perform failure-case analysis across varying haze densities.

10. Conclusion

This project implemented a complete single-image haze-removal restoration pipeline using the Dark Channel Prior together with a manually implemented guided filter for transmission-map refinement, built entirely in Python with OpenCV, NumPy, Matplotlib and scikit-image. A clean outdoor photograph was used to synthesise a controlled hazy version via the atmospheric scattering model, enabling objective, ground-truth-based evaluation. The pipeline estimated the atmospheric light and transmission map directly from the hazy image, refined the transmission map to be edge-aware, and inverted the scattering model to recover the scene radiance. The restored image showed a clear visual and quantitative improvement over the hazy input, with PSNR rising from 12.71 dB to 17.76 dB and SSIM rising from 0.766 to 0.924. The project demonstrated that a classical, prior-based restoration technique — without any training data — can meaningfully recover contrast, colour and structural detail lost to atmospheric haze, while also revealing the known limitations of the Dark Channel Prior (sky-region artefacts, fixed atmospheric-light assumption) that motivate the future improvements outlined above.